

Arsum Ullah

Data Science & Machine Learning Student | Data Structures Lab Demonstrator

ullaharsum@gmail.com linkedin.com/in/arsum-ullah-a11245292/ github.com/Shell-aurum

TECHNICAL SKILLS

Languages & Core Concepts: C++, x86 Assembly, Python, Linear Algebra, Probability Theory, Neural Networks

Infrastructure & Automation: n8n (Self-hosted), Docker, AI Agent Workflows

Prototyping: Figma

EXPERIENCE

Data Structures Lab Demonstrator

[Fast Nuces Islamabad]

[Feb 2026] – Present

[Islamabad, Pakistan]

- Evaluate student implementations of fundamental data structures including linked lists, binary search trees, AVL trees, stacks, and queues, enforcing algorithmic efficiency and correctness under defined time and space constraints.
- Debug complex C++ logic for students, diagnosing pointer management errors, memory leaks, and off-by-one conditions in recursive and iterative solutions.
- Deliver technical explanations of Big-O complexity analysis to cohorts of [X] students per session, reinforcing industry-standard coding practices.

PROJECTS

Automated Content Pipeline

[2026]

- Designed and deployed a hands-free content curation system using self-hosted n8n orchestrated within Docker containers, eliminating manual intervention across the full pipeline.
- Engineered multi-step AI agent workflows that automate content sourcing, filtering, and scheduling, demonstrating scalable backend logic and production-grade API orchestration.
- Reduced manual processing time by [X]% through event-driven triggers and automated routing between external services.

High-Performance Movie Data Manager

[2025]

- Developed a movie database management system in C++, implementing self-balancing AVL trees to achieve $O(\log n)$ search and insertion operations across datasets of [X]+ records.
- Built custom doubly linked lists for dynamic record management, replacing static array structures and improving memory utilisation and deletion efficiency.
- Validated correctness of tree rotations and pointer logic through systematic unit testing, ensuring zero data corruption on edge-case inputs.

Autonomous Swarm Robotics – Feasibility Study & Architecture

[2026]

- Conducted a full feasibility study for a five-unit autonomous swarm system built on 2WD chassis platforms, evaluating sensor integration, communication protocols, and inter-agent coordination strategies.
- Produced a hardware-software integration architecture defining control logic layers, obstacle avoidance algorithms, and decentralised decision-making patterns for the swarm.
- Identified critical technical constraints related to real-time processing and power budgeting, directly informing component selection for the prototype phase.

Career Lens – Interactive Application Prototype

[2026]

- Mapped complete user flow and interface requirements in Figma prior to any backend implementation, defining screen states, navigation logic, and component hierarchy for a career-guidance application.
- Translated user research into low- and high-fidelity wireframes, reducing design iteration cycles and providing a clear technical specification for development handoff.

EDUCATION

Bachelor of Science in Computer Science

[2028]

[Fast Nuces Isb]

[Islamabad, Pakistan]

CERTIFICATIONS

- **Machine Learning Specialization** – DeepLearning.ai | Coursera *In Progress, [2026]*
Rigorous industry-standard curriculum covering supervised learning, advanced algorithms, and neural network foundations.